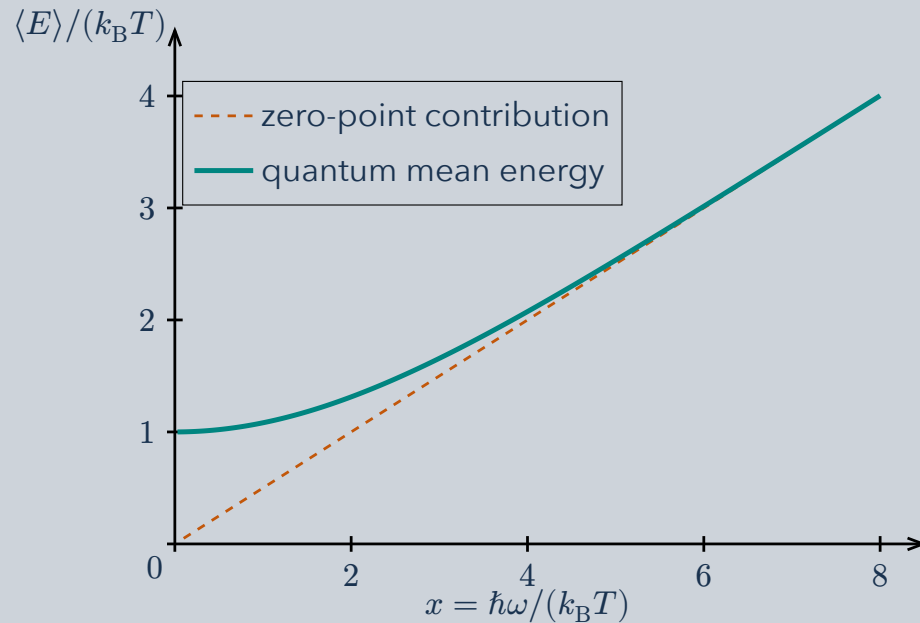


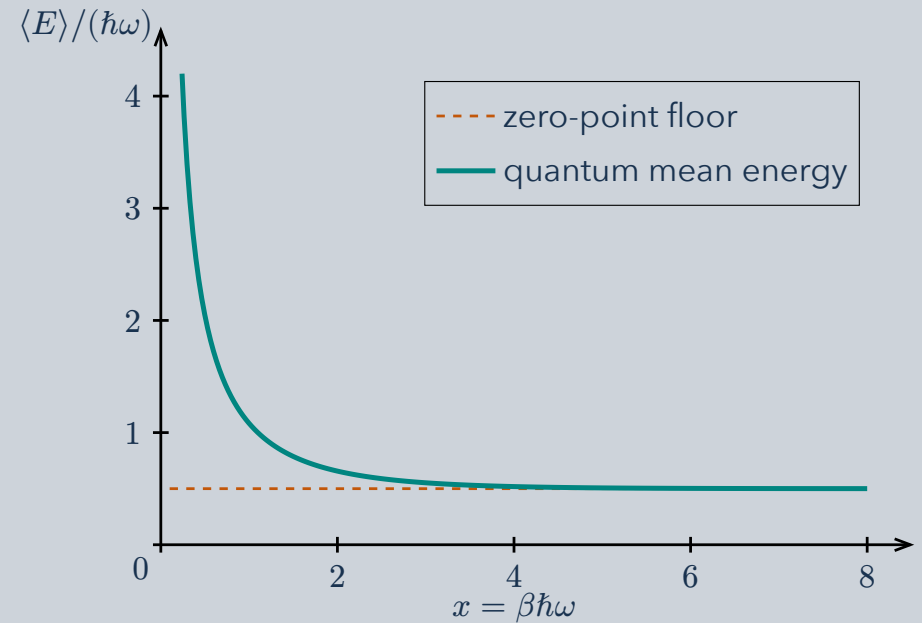
One energy formula connects thermal motion with zero-point motion. Compare the level spacing $\hbar\omega$ with the thermal energy $k_{\text{B}}T$.

1 Increase frequency at fixed temperature



At small level spacing, $\langle E \rangle \approx k_{\text{B}}T$: classical equipartition. At large spacing the zero-point term dominates. The energy rises monotonically.

2 Cool at fixed frequency



Move right by increasing $\beta = 1/(k_{\text{B}}T)$. Thermal excitations freeze out; the mean energy approaches $\hbar\omega/2$, rather than zero.

$\langle E \rangle = \hbar\omega \left(\frac{1}{2} + \frac{1}{e^{\beta\hbar\omega} - 1} \right)$. **The same horizontal variable, two energy units.** ω is angular frequency; $\hbar$ is the reduced Planck constant.